

MITIGATOR

Product Requirements Document

Stocklana 2026 Hackathon • Version 1.0 • September 2026

AI-powered pre-trade intelligence, risk mitigation, social intelligence, and execution for tokenized stocks on Solana

Core product loop

Discover → Research → Verify → Analyze → Score Risk → Simulate → Mitigate → Compare Execution → Execute → Monitor → Discuss → Learn

Build objective: make the judges understand the product in 30 seconds, trust the data in 2 minutes, and witness a complete intelligence-to-execution flow in under 5 minutes.

1. Executive Summary

MITIGATOR is a Solana-native trading intelligence platform for tokenized stocks. Its central promise is simple: before a user buys or sells a tokenized stock, MITIGATOR combines verified market, fundamental, news, social, event, portfolio, token, oracle, and onchain-liquidity data to explain the opportunity and risk, recommend mitigation actions, compare available execution venues, and let the user execute with explicit guardrails.

The product is intentionally broader than a stock screener. It is a complete intelligence-to-execution loop: Discover → Research → Verify → Analyze → Score Risk → Simulate → Mitigate → Compare Execution → Execute → Monitor → Discuss → Learn. All planned features remain in scope: AI trading, AI copilot, paper trading, portfolio/risk management, recurring buys and DCA, copy trading/strategies, social stock timelines, alerts, robo portfolios, market watch, long-range charts, cross-venue execution, cross-chain expansion, and autonomous-agent workflows.

The key product distinction is that MITIGATOR does not treat AI as a stock-price oracle. AI is the reasoning and explanation layer over structured, provenance-aware data. Every important claim shown to a user should be traceable to a source, timestamp, confidence level, and—where possible—corroborating sources.

2. Product Vision

Vision: become the risk-aware intelligence and execution layer for the tokenized-equity economy.

Mission: make every tokenized-stock trade explainable, comparable, verifiable, and risk-aware before a wallet signature is requested.

Core positioning: “Don't just trade. Understand the risk first.”

Secondary product line: “Research. Risk. Route. Execute.”

3. Problem Statement

Tokenized stocks make equity exposure composable and tradable onchain, but they introduce a combined information problem. A trader must understand the underlying company and market while also evaluating token integrity, price reference, corporate-action mechanics, liquidity, spread, price impact, venue quality, and portfolio concentration.

Existing brokerage-style experiences generally optimize for access and execution. Research terminals optimize for information. DeFi interfaces optimize for swaps. MITIGATOR combines these layers into one pre-trade decision workflow and makes the reasoning visible.

The product must avoid the false assumption that “more data automatically means more reliable data.” Reliability comes from source quality, freshness, independence, corroboration, conflict detection, provenance, and confidence-weighted fusion.

4. Target Users

Primary: crypto-native retail investors who want exposure to tokenized equities but do not want to manually research financial news, filings, liquidity, token mechanics, and execution venues.

Secondary: active traders who want fast multi-source intelligence, venue comparison, alerts, and guarded execution.

Advanced: strategy builders, copy traders, quant/AI users, and portfolio managers who want programmable research, simulation, and agent-assisted execution.

Future: DAOs, fintech apps, wallets, brokers, and institutional users who want MITIGATOR's risk/intelligence APIs.

5. Product Principles

Evidence before opinion — AI claims must be grounded in retrieved data.

Source provenance everywhere — users can inspect where a signal came from.

Risk before execution — the system analyzes the proposed order, not just the asset.

Solana-native by design — wallet, onchain state, liquidity, routing and transaction verification are first-class.

Explainability over black-box predictions — show the factors behind every score.

Human approval for live execution — autonomous recommendations can be simulated; live trades require explicit user authorization/signature.

Fast path for experts, guided path for beginners — progressive disclosure.

3D where it creates attention or comprehension; conventional charts and tables where precision matters.

Never conflate underlying-equity history with token/onchain history.

6. Full Feature Scope

- A. Market Intelligence: asset discovery, watchlists, live prices, historical market watch, technical indicators, fundamentals, earnings, analyst/consensus data where licensed, macro indicators, sector comparisons, volatility, correlation, drawdown, valuation, and market status.
- B. AI Intelligence: AI pre-trade brief, bullish/bearish thesis, risk explanation, catalyst detection, “why not?” analysis, scenario analysis, portfolio-aware recommendations, natural-language copilot, trade-plan generation, strategy construction, autonomous monitoring, and agent workflows.
- C. Risk & Mitigation: MITIGATOR Score, market risk, event risk, news risk, liquidity risk, onchain/token risk, portfolio concentration risk, correlation risk, execution risk, order-size simulation, slippage limits, DCA plans, position-size suggestions, stop/alert policies, and policy-based trade blocking.
- D. Execution: multi-venue price comparison, Jupiter routing, xChange RFQ where onboarding/access permits, Raydium/Orca/Meteora liquidity discovery, quote freshness, spread, expected slippage, price impact, route quality, wallet signing, transaction confirmation, execution receipts, and post-trade monitoring.
- E. Social Intelligence: stock-specific public timeline, user posts, sentiment, verified-account indicators, source reputation, citations, X integration, Reddit/Stocktwits/other permitted public sources, discussion threads, reactions, spam controls, misinformation flags, and AI summaries.
- F. Social Trading: strategy profiles, trader/strategy performance, risk-adjusted metrics, transparent holdings, copy/follow strategies, copy-size adjustment based on the user's own risk limits, and paper-copy mode.
- G. Portfolio: wallet portfolio, cost basis, P&L, allocation, sector exposure, correlation, drawdown, risk budget, recurring buys, DCA, robo portfolios, alerts, rebalancing suggestions, and simulation.
- H. Market Watch: multi-year charts, 1D/1W/1M/3M/6M/YTD/1Y/5Y/MAX, candlesticks, volume, volatility, drawdowns, event/news overlays, sentiment overlays, MITIGATOR Score history, compare assets, normalized performance, and underlying-vs-token history labeling.
- I. Cross-chain: Solana first; architecture supports future Ethereum, Arbitrum, Base, TON and other supported tokenized-stock networks without redesigning the domain model.
- J. Alerts: price, news, sentiment, risk-score, liquidity, price-divergence, corporate-action, earnings, portfolio concentration, strategy, and custom natural-language alerts.
- K. Education & Trust: source-quality badges, data freshness, conflict indicators, token verification, issuer information, risk disclosures, methodology pages, and audit-friendly event logs.

7. Core User Journey

1. User connects a Solana wallet or starts in read-only/paper mode.
2. User searches for an asset such as NVDAx, AAPLx, SPYx, or TSLAx.
3. MITIGATOR resolves the canonical asset, issuer, underlying ticker, Solana mint/deployment, oracle/reference sources, market status, and corporate-action/multiplier state.
4. User sees a 3D asset hero plus a conventional precision chart and MITIGATOR Score.
5. User opens Intelligence to inspect news, filings, fundamentals, social discussion, catalysts, and the AI thesis.
6. User opens Risk to see factor-level scoring and evidence.
7. User enters a proposed order. MITIGATOR calculates portfolio impact, liquidity impact, expected slippage, venue availability, event risk, and recommended mitigation.
8. User chooses a mitigation plan or overrides it with explicit acknowledgement.
9. MITIGATOR compares execution venues and produces a quote/route.
10. User signs the transaction. The app verifies confirmation and stores an execution receipt.
11. Portfolio and alerts update. The stock's social timeline and user's activity can continue the discussion.

8. MITIGATOR Score & Risk Engine

The MITIGATOR Score is a dynamic, explainable score from 0–100. Higher is better for overall trade readiness; a separate risk severity scale is shown so users are not forced to interpret one number.

Recommended factor model: Market Quality 15%; Fundamental/Business 15%; News & Sentiment 15%; Event/Catalyst Risk 10%; Liquidity & Execution 15%; Token/Onchain Integrity 15%; Portfolio Fit 10%; Data Confidence 5%.

The engine must expose both the score and its evidence. Each factor contains: value, normalized score, direction, source list, timestamp, freshness, confidence, and explanation.

Trade-level risk must differ from asset-level risk. A liquid stock can become a high-risk trade if the user's proposed order is too large, creates concentration, crosses a major event, or encounters poor execution conditions.

Risk states: LOW, MODERATE, ELEVATED, HIGH, CRITICAL. The exact thresholds should be configurable and versioned.

Every scoring-model change must be versioned so historical scores can be reproduced.

9. AI System

The AI layer consumes structured evidence packets rather than raw untrusted web text. Each packet contains source ID, source tier, publication timestamp, retrieval timestamp, content hash/reference, ticker/entity mapping, sentiment/event tags, and confidence.

AI functions: Research Agent, News Agent, Filing Agent, Social Agent, Risk Agent, Execution Agent, Portfolio Agent, Strategy Agent, Alert Agent, and Orchestrator/Copilot.

AI output schema: summary, thesis, counter-thesis, evidence, uncertainty, catalysts, invalidation conditions, risk flags, mitigation actions, execution recommendation, and citations.

The AI must not invent prices, filings, events, quotes, liquidity, or trade confirmations. If evidence is missing or conflicting, it must say so.

Live autonomous trading is gated by policy: max order value, max daily loss, max position size, allowed assets, allowed venues, max slippage, minimum data confidence, and event blackout windows.

10. Verified Data & Source Architecture

MITIGATOR will use a source registry rather than a loose collection of scrapers. Every provider is registered with: provider name, category, legal/access method, API endpoint, authentication mode, update cadence, fields, historical depth, license/terms status, reliability tier, failover provider, and validation rules.

Tier 0 — canonical/onchain/issuer: xStocks official API and documentation; Solana RPC/onchain token metadata; Pyth price feeds; issuer proof-of-reserves and multiplier data where available.

Tier 1 — primary authoritative financial sources: SEC EDGAR filings and XBRL; company investor-relations releases; official exchange/market-status data; Federal Reserve/FRED and other official macroeconomic releases.

Tier 2 — licensed/structured aggregators: Finnhub, Alpha Vantage, NewsAPI and other providers selected after checking entitlement, licensing, rate limits, historical depth and permitted use.

Tier 3 — public social and alternative sources: X API, Reddit API, Stocktwits or other permitted feeds, public social pages, blogs, forums and media. These are signal sources, not truth sources.

Social/news scraping is allowed only where the source's terms, robots/access controls, licensing and applicable law permit it. Prefer official APIs or licensed feeds. MITIGATOR must preserve the original source URL/ID and never present social chatter as verified fact.

Source fusion: corroborate material claims across independent sources. The system should down-weight duplicate syndications and detect conflicting reports. A claim repeated by 30 sites that all copied one article is still one information source.

Freshness and conflict rules: stale quotes are rejected for execution; stale news is labeled; conflicting high-impact claims create an "Unconfirmed/Conflicting" state; critical execution data requires a current venue/quote response.

11. Recommended Data Provider Matrix

xStocks — canonical tokenized-stock asset metadata, asset price data, oracle information, multiplier/corporate-action state, proof-of-reserves information, and xChange RFQ/availability where access is enabled.

Pyth — independent price-reference/oracle layer and price confidence/staleness checks where the relevant feed exists.

Solana RPC — token/account/transaction state and confirmation evidence.

SEC EDGAR — filings, XBRL financial data, insider/ownership filings and primary regulatory evidence.

FRED — macroeconomic time series and release calendar context.

Alpha Vantage or Finnhub — long-range historical OHLCV, indicators, fundamentals/news/alternative data subject to plan/licensing.

NewsAPI/GDELT — broad news discovery and historical/global coverage; use source scoring and corroboration rather than treating all publishers equally.

X API — real-time/public posts and account/post discovery subject to API access and pricing. Use official API access rather than unauthorized scraping where possible.

Raydium, Orca, Meteora — onchain pool/liquidity/quote information where supported; verify pool identity and token mint before using data.

Jupiter — routing/quote/execution layer where supported for the target assets.

xChange — issuer-side atomic RFQ path where onboarding and API credentials permit it.

12. Data Verification & Provenance Requirements

Every displayed data point must carry an internal provenance object: provider, endpoint, retrieved_at, published_at (if applicable), asset ID, source tier, freshness window, confidence, and checksum/content reference where practical.

Canonical entity resolution: map company ticker → underlying security → token symbol → Solana mint → issuer → oracle feed(s) → supported venues. Never rely on ticker text alone.

Price verification: compare token/venue price against independent reference data; calculate deviation; flag abnormal divergence; never use a wrapped-token accounting exchange rate as a market price.

News verification: classify primary vs secondary reporting, detect duplicates, extract claims, and require corroboration for high-impact claims before raising a high-confidence alert.

Social verification: label posts as opinion, reported claim, official statement, or unverified chatter. Official company/issuer/regulator accounts receive higher source authority but are still timestamped and attributed.

Execution verification: quote must be fresh, route must be reproducible, expected output must be shown, slippage/price impact must be calculated, and transaction confirmation must be independently verified onchain.

13. Stock Market Timeline & Social Timeline

Every stock page includes a unified timeline with two synchronized modes: Market Timeline and Community Timeline.

Market Timeline overlays price/volume with earnings, filings, dividends, splits, major news, macro events, risk-score changes, liquidity events, and notable onchain deviations.

Community Timeline contains user thoughts, questions, trade theses, replies, reactions, sentiment, and linked evidence. Posts can be attached to an asset and optionally to a time window or event.

Users can filter by Verified Sources, News, Filings, Company, Analysts, Community, AI Signals, and Onchain Events.

A post can be marked as “Evidence attached” only when it references a source. Users may still publish opinions without evidence, but the UI clearly distinguishes opinion from verified information.

AI summarizes the conversation periodically: consensus, disagreement, emerging narratives, misinformation/low-quality clusters, and notable changes in sentiment.

Anti-abuse: rate limits, duplicate-content detection, spam scoring, report/block tools, moderation queue, and clear financial-risk disclosures.

14. Historical Market Watch & Charts

The Market Watch experience must support long-range analysis: 1D, 1W, 1M, 3M, 6M, YTD, 1Y, 5Y and MAX where source coverage exists.

Charts: candlestick/line, volume, relative performance, drawdown, volatility, moving averages, RSI/MACD where appropriate, correlation, and normalized performance comparisons.

Event overlays: earnings, filings, dividends, splits, major news, macro releases, and AI-detected regime changes.

Dual history model: Underlying Equity History and Token/Onchain History are separate series. If the token launched later than the underlying security, the UI explicitly labels the earlier history as underlying-equity context.

The user can compare up to 5 assets and normalize them to 100 at a selected start date.

15. Multi-Venue Execution

Execution screen compares available routes using expected received amount, quoted price, spread, price impact, estimated slippage, liquidity/depth, fees, quote age, route complexity, and settlement characteristics.

Jupiter is the default routing layer for supported swaps. Raydium/Orca/Meteora can contribute venue/liquidity information and direct execution where appropriate. xChange is used for issuer RFQ where available and authorized.

The UI must distinguish indicative/reference price from executable quote.

Quote lifecycle: request → validate → display expiry → user confirmation → wallet signature → broadcast → confirmation → receipt.

Guardrails: max slippage, max price impact, minimum received, max order value, approved assets, approved venues, and user-specific risk policy.

16. AI Trading, Paper Trading & Autonomous Agents

Paper Trading is a first-class environment with simulated wallet balances, realistic quote/fee/slippage assumptions, trade journal, P&L, drawdown, win rate, exposure and risk-adjusted performance.

AI Trading Agent can scan eligible assets, rank opportunities, construct a proposed portfolio, simulate trades, and monitor positions. Live execution is disabled by default until the user explicitly enables a policy.

Robo Portfolios support configurable objectives such as Conservative, Balanced, Growth, Momentum, Dividend/Income where data supports the strategy, and Custom.

Recurring buys/DCA allow schedules, amount, target assets, maximum price/slippage, event blackout windows, and automatic pause conditions.

Autonomous policies include kill switch, daily loss cap, position cap, maximum trade count, data-confidence minimum, and mandatory human approval for defined risk classes.

17. Copy Trading & Strategy Marketplace

Users can create or follow strategies. A strategy profile shows transparent historical performance, drawdown, volatility, risk score, asset concentration, holding period, trade frequency, and methodology.

Copy trading is risk-aware rather than blind. MITIGATOR calculates the user's recommended copy size based on available capital, current exposure, correlation and personal risk policy.

Strategy actions can be simulated before following live.

Strategy provenance: every performance number must be generated from recorded fills/quotes, not self-reported claims.

Leaderboards rank by risk-adjusted performance, consistency and transparency—not raw ROI alone.

18. Portfolio & Risk Management

Portfolio dashboard: total value, cash/stablecoin, holdings, P&L, realized/unrealized P&L, allocation, sector exposure, correlation, concentration, drawdown, risk budget, and upcoming catalysts.

Position risk cards show current score, thesis, invalidation conditions, news/event changes, liquidity, and recommended actions.

Scenario simulator: "What happens if NVDA drops 10%?", "What if earnings miss?", "What if liquidity falls 40%?"

Rebalancing recommendations must show before/after allocation and expected risk change.

Alerts can trigger on price, drawdown, concentration, score deterioration, liquidity, news, sentiment, and event proximity.

19. 3D / Modern UI Direction

Visual goal: a premium, cinematic, futuristic financial interface that immediately distinguishes MITIGATOR from generic dashboards while retaining institutional-grade information density.

Hero: interactive 3D asset visualization using React Three Fiber/WebGL or CSS 3D. A tokenized-stock “orb”/glass object can display price, score and live market state. Keep 3D decorative and interactive—not the only way to access information.

Market overview: layered glass panels, depth, soft bloom, subtle particle/market-grid background, animated price pulses, floating data cards, and responsive micro-interactions.

Risk visualization: 3D radial/ring visualization for factor health plus conventional numeric breakdown. Use motion to reveal changes rather than constant animation.

Stock page: 3D hero + precision chart + intelligence panels + timeline. The chart remains conventional so exact prices and dates are readable.

Color language: near-black/deep graphite base, electric green for healthy/positive states, amber for caution, red for critical, cool violet/cyan as restrained accent colors. Avoid neon overload.

Motion: 150–250ms UI transitions, spring-based card movement, chart transitions, wallet state animations, and a strong but short hero entrance. Respect prefers-reduced-motion.

Performance: lazy-load WebGL, use instancing where needed, cap DPR, avoid heavy 3D on mobile, provide 2D fallback, and target fast first contentful render.

Design principle: “3D for attention; 2D for decisions.”

20. Primary Screens

- Landing / cinematic product hero.
- Discover / global tokenized-stock market.
- Market Watch / multi-asset dashboard.
- Asset Search / command palette.
- Stock Detail / 3D hero, chart, intelligence, risk, timeline, execution.
- AI Copilot / persistent contextual assistant.
- Pre-Trade Analysis / MITIGATOR brief.
- Trade Simulator / scenario and order-impact analysis.
- Execution Router / venue comparison and quote.
- Wallet / positions and transaction history.
- Portfolio / allocation and risk.
- Alerts Center.
- Community / stock-specific social timeline.
- Strategy Marketplace / copy trading.
- Paper Trading / simulated portfolio.
- Robo Portfolio Builder.
- AI Agent Console / autonomous policies and monitoring.
- Data Provenance / source inspector.
- Settings / risk policy, data preferences, API/provider status.

21. Suggested Frontend Architecture

React + TypeScript + Vite. Tailwind CSS + shadcn/ui for the conventional interface. React Three Fiber/Three.js for the 3D layer. Recharts or Lightweight Charts for financial charts. TanStack Query for server state. Zustand for lightweight client state. Wallet Adapter for Solana wallet connection.

Route structure: /, /discover, /markets, /asset/:symbol, /trade/:symbol, /portfolio, /paper, /strategies, /community, /alerts, /agent, /settings.

Component domains: market, asset, intelligence, risk, execution, portfolio, social, strategy, agent, provenance, 3d.

Design system tokens must be centralized so the visual language can be tuned rapidly during the hackathon.

22. Backend & Data Architecture

Node.js + TypeScript backend with a modular service architecture. Recommended modules: asset-service, market-data-service, news-service, filing-service, social-service, sentiment-service, risk-service, portfolio-service, execution-service, alert-service, strategy-service, ai-orchestrator, provenance-service.

PostgreSQL/Supabase is the system of record. Redis can be added for quotes, rate limits and short-lived jobs. A queue/worker layer handles ingestion, deduplication, enrichment and AI processing.

Event-driven pipeline: provider ingestion → normalization → entity resolution → validation → enrichment → scoring → storage → API/WebSocket → UI.

All provider adapters implement the same internal interface so providers can be swapped without changing the UI or risk engine.

23. Core Data Model

users: id, profile, preferences, risk_policy.

wallets: user_id, chain, address, label, verified_at.

assets: id, symbol, name, underlying_symbol, issuer, chain, mint, token_standard, status.

asset_deployments: asset_id, chain, mint/address, decimals, multiplier, verified_at.

providers: id, name, tier, category, access_method, terms_status, reliability_score.

market_quotes: asset_id, provider_id, bid, ask, last, volume, timestamp, freshness, confidence.

historical_bars: asset_id/underlying_id, timeframe, OHLCV, source, timestamp.

news_items: source, URL/ID, title, body_hash, published_at, retrieved_at, entities, sentiment, credibility.

filings: company, CIK, form, accession, filed_at, period, extracted_metrics, URL.

social_posts: platform, author, post_id, asset_id, text_hash, published_at, sentiment, source_class, credibility.

risk_scores: asset_id, trade_id, model_version, factor_scores, total_score, confidence, evidence_refs, created_at.

quotes/executions: venue, route, quote_id, amount, expected_output, slippage, price_impact, expiry, tx_signature, status.

strategies, strategy_positions, followers, paper_trades, alerts, timeline_events, ai_runs, provenance_records.

24. API Surface

- GET /assets, /assets/:symbol, /assets/:symbol/market, /assets/:symbol/history
- GET /assets/:symbol/news, /filings, /social, /events, /risk, /provenance

- POST /ai/analyze, /ai/simulate, /ai/portfolio, /ai/strategy
- POST /trade/quote, /trade/simulate, /trade/execute
- GET /execution/venues/:symbol
- GET /portfolio, /portfolio/risk, /portfolio/scenarios
- GET/POST /strategies, /strategies/:id/follow, /strategies/:id/simulate
- GET/POST /alerts
- GET/POST /timeline/:assetId/posts
- GET /providers/status, /providers/:id/provenance
- WebSocket/SSE channels for prices, alerts, AI signals, execution status, and community updates.

25. Security, Safety & Trust

Never store private keys. Transactions are signed by the user's wallet. Server-side signing is prohibited for the consumer app unless a future institutional custody product explicitly adds it.

Use strict allowlists for executable asset mints and venue programs. Verify program IDs and token mints before constructing transactions.

AI has no direct authority to bypass risk policy. The policy engine sits between AI recommendations and execution.

Rate-limit social ingestion and user posting. Sanitize all user content.

Protect API keys server-side; never expose provider secrets in the frontend.

Log all live trade decisions: user request, data snapshot IDs, model version, risk score, mitigation, quote, user approval, wallet signature, transaction confirmation.

Display financial-risk disclaimers and clearly distinguish information from personalized financial advice.

26. Non-Functional Requirements

Performance: primary dashboard interactive as early as possible; lazy-load 3D and heavy charts.

Reliability: graceful provider degradation; a single failed provider must not take down the asset page.

Data freshness: live quote paths must display quote age and reject stale execution data.

Observability: provider health, ingestion lag, AI latency, WebSocket health, quote failure rate, and transaction success rate.

Accessibility: keyboard navigation, semantic labels, reduced-motion support, readable contrast.

Responsive: desktop-first for hackathon judging, but functional mobile layout for core asset/risk/execute flows.

Auditability: every score and AI answer can be traced to evidence and model version.

27. Hackathon Acceptance Criteria

- A judge can connect a Solana wallet, search an xStock, inspect verified asset metadata, see a multi-year market chart, read a source-backed AI brief, inspect a stock-specific community timeline, enter a trade size, see portfolio and execution risk, compare available execution paths, apply mitigation, sign a real or test transaction, and see the resulting position.
- A judge can ask the AI copilot a natural-language question and receive an answer containing evidence/source references and uncertainty.
- A judge can inspect the data provenance panel and understand which source produced each major data point.

- A judge can switch to paper trading and simulate a strategy without risking funds.
- A judge can view copy/strategy performance and see how MITIGATOR adjusts the recommended copy size to the user's portfolio risk.
- The application remains visually polished under the full feature set, with 3D used as an attention layer rather than a usability obstacle.

28. Seven-Day Delivery Plan

- Day 1 — Foundation: React/TS/Vite, design system, routes, wallet, database, asset registry, provider registry, xStocks ingestion, canonical entity mapping.
- Day 2 — Market & Asset Intelligence: live quotes, historical market-watch charts, asset detail, underlying-vs-token history, filings/news ingestion, source provenance UI.
- Day 3 — MITIGATOR Engine: risk factors, scoring model, evidence packets, AI research brief, counter-thesis, event risk, portfolio-aware order simulation.
- Day 4 — Execution: Jupiter routing, xChange integration path if access is available, venue comparison, quote freshness, slippage/price-impact calculator, wallet signing, transaction confirmation.
- Day 5 — Social & Portfolio: stock timelines, community posts, sentiment aggregation, portfolio dashboard, alerts, paper trading, trade journal.
- Day 6 — Advanced Intelligence: AI copilot, strategy marketplace, risk-aware copy trading, DCA/recurring buys, robo portfolios, autonomous policy console, scenario simulator.
- Day 7 — Killer Experience: 3D hero, motion polish, provenance explorer, error states, responsive pass, demo seed data, end-to-end QA, submission video, pitch deck, live demo rehearsal.

29. Demo Script

- Opening: "Buying a tokenized stock takes seconds. Knowing whether you should buy it—and how to buy it safely—is the hard part."
- Search NVDAx. Show the 3D asset hero and MITIGATOR Score.
- Open Intelligence. Show recent news, a filing, social conversation, and upcoming event. Click a source to prove provenance.
- Ask: "Should I buy \$2,000 of NVDAx?" AI explains the thesis, risk, and counter-thesis.
- Open trade simulator. Show portfolio concentration, event risk, liquidity, and estimated price impact.
- MITIGATOR recommends a smaller first order plus DCA. Click Apply Mitigation.
- Open execution comparison. Show Jupiter/xChange/other supported venues with quote age and expected output.
- Sign the transaction. Confirm on Solana. Show the updated portfolio and alert creation.
- Finish with the community timeline and ask the copilot: "What changed around NVDAx today?"
- Closing: "MITIGATOR doesn't tell you what to blindly buy. It tells you what you're buying, what can go wrong, how to mitigate it, and where to execute."

30. Success Metrics

- Activation: wallet connection and first asset analysis.
- Research depth: percentage of asset sessions that open risk, provenance, or intelligence views.
- Trade readiness: percentage of simulated trades with a completed risk/mitigation review.

- Execution quality: quote-to-execution success rate, average price impact, average slippage vs benchmark.
- AI quality: grounded-answer rate, citation coverage, unsupported-claim rate, user feedback.
- Data quality: source freshness, provider uptime, conflict detection, duplicate-news rate.
- Community: posts per active asset, evidence-attached post ratio, engagement, moderation rate.
- Strategy: paper-copy adoption, risk-adjusted performance, drawdown, and strategy transparency.
- Hackathon: time-to-understand for a new judge, end-to-end demo completion rate, and visual/UX evaluation.

31. Risks & Mitigations

- Data licensing risk → use official APIs/licensed feeds, maintain provider terms registry, and isolate adapters.
- Social misinformation → source classes, credibility scoring, corroboration, conflict labels, and evidence links.
- AI hallucination → structured evidence packets, citations, schema validation, confidence thresholds, and no execution authority.
- Liquidity manipulation → independent reference price, pool-quality checks, abnormal-volume detection, minimum-liquidity policies.
- Token/corporate-action errors → canonical asset mapping and multiplier-aware accounting. Never assume raw Solana balance equals displayed economic balance for xStocks.
- Execution failure → quote expiry, transaction simulation where supported, retry/requote flow, and clear user status.
- Over-complexity → progressive disclosure: the judge sees a simple score first, then can drill into the underlying evidence.
- 3D performance → lazy loading, reduced-motion mode, 2D fallback, and strict GPU budgets.

32. Definition of Done

- MITIGATOR is “done” for the Stocklana submission when the complete primary journey works end-to-end on Solana; all major data claims have provenance; the AI can explain its reasoning with evidence; the risk engine can analyze a proposed trade; at least one real execution path can be demonstrated; paper trading works; social timeline and multi-year charts are functional; the portfolio updates after execution; and the 3D interface is polished enough to create immediate visual differentiation without sacrificing usability.

33. Primary Research Sources

xStocks Developer Documentation

<https://docs.xstocks.fi/developers>

Canonical developer source for asset data, price data, multipliers, oracle information and xChange.

xStocks xChange Integration

<https://docs.xstocks.fi/developers/xchange-atomic-rfq>

Atomic RFQ availability and Solana execution flow.

xStocks Multipliers

<https://docs.xstocks.fi/developers/multipliers>

Corporate-action multiplier mechanics.

xStocks Product / How It Works

<https://docs.xstocks.fi/docs/how-xstocks-work>

Tokenized equity structure and primary/secondary market model.

Pyth Price Feeds

<https://docs.pyth.network/price-feeds>

Independent price-feed architecture and provider coverage.

Solana

<https://solana.com/>

Solana capital-markets and network context.

Jupiter Developers

<https://dev.jup.ag/>

Solana routing/swap developer documentation.

Raydium Docs

<https://docs.raydium.io/>

Solana liquidity, SDK and API documentation.

Orca Whirlpools SDK

<https://dev.orca.so/developers/sdks/overview>

Solana concentrated-liquidity SDK documentation.

SEC EDGAR APIs

<https://www.sec.gov/search-filings/edgar-application-programming-interfaces>

Primary regulatory filing and XBRL APIs.

FRED

<https://fred.stlouisfed.org/>

Official macroeconomic data.

Alpha Vantage API

<https://www.alphavantage.co/documentation/>

Historical market data, fundamentals and market-intelligence API options.

Finnhub

<https://finnhub.io/>

Market, fundamentals and alternative-data API options.

NewsAPI

<https://newsapi.org/>

Broad structured news search API.

GDELT

<https://www.gdeltproject.org/>

Large-scale global news/event dataset and analysis tools.

X Developer Platform

<https://docs.x.com/overview>

Official X data/API access.